EP2108 - Special Theory of Relativity

Assignment 3: Tensors

Part A: Cartesian tensors and index notation

1. Under a rotation $x'_i = a_i^j x_j$ with $a_i^j a_j^k = \delta_i^k$, a rank-2 Cartesian tensor transforms as $T'_{ij} = a_i^k a_j^l T_{kl}$.

(a) Show that δ_i^j has the same components in every rotated frame.

(b) Show that the trace T_{ii} is a scalar.

2. Verify the identity $\varepsilon_{ijk}\varepsilon_{lmk} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$, then use it to show $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$.

3. Under an orthogonal transformation $x'_i = a_i^{\ j}\, x_j$, where $a_i^{\ j}\, a_j^{\ k} = \delta_i^{\ k}$ and $a_j^{\ k} = a_k^{\ j}$ (so $\det a = \pm 1$), a rank-3 Cartesian tensor transforms as

$$T'_{ijk} = a_i^{\ l}\, a_j^{\ m}\, a_k^{\ n}\, T_{lmn}.$$

Show that ε_{ijk}, which has the same numerical values $(0, \pm 1)$ in every frame, is a rank-3 *pseudotensor*: under an improper transformation ($\det a = -1$) it picks up an extra sign relative to this transformation law. Explain why $\mathbf{A} \times \mathbf{B}$ is then a pseudovector.

4. Show that any rank-2 tensor splits as $T_{ij} = S_{ij} + A_{ij}$ with S symmetric and A antisymmetric, and that this split is preserved under rotations. Show that $S_{ij}A_{ij} = 0$.

Part B: Quotient theorem and tensor properties

5. State the quotient theorem. Prove that if $A_{ij}B^j = C_i$, with B^j an arbitrary vector and C_i a vector, then A_{ij} is a rank-2 tensor, i.e. $A'_{ij} = a_i^{\ k}\, a_j^{\ l}\, A_{kl}$ under $x'_i = a_i^{\ j} x_j$.

6. The moment-of-inertia tensor is $I_{ij} = \int \rho\,(r^2\delta_{ij} - x_i x_j)\, dV$. Show that it is a symmetric rank-2 tensor, and find the principal axes and moments of inertia of a uniform solid cube of side a about its centre (compute I_{ij} in axes parallel to its edges).

Part C: Curvilinear coordinates and the metric

7. In general coordinates q^i, the metric is $g_{ij} = \dfrac{\partial x^k}{\partial q^i}\dfrac{\partial x^k}{\partial q^j}$, so $ds^2 = g_{ij}\, dq^i dq^j$.

(a) Find g_{ij} and g^{ij} for plane polar coordinates (r, θ).

(b) Find g_{ij} for spherical coordinates (r, θ, φ).

8. Distinguish the contravariant components A^i of a vector from its covariant components $A_i = g_{ij}A^j$ by showing how each transforms under a change of coordinates $q^i \to q'^i$. For $\mathbf{A} = r\,\hat{\mathbf{r}}$ in plane polar coordinates (r, θ), find A^i and A_i, and verify that $A_i A^i = A^2$.

9. The Christoffel symbols are $\Gamma^k_{ij} = \frac{1}{2}g^{kl}\,(\partial_i g_{lj} + \partial_j g_{li} - \partial_l g_{ij})$. Compute all non-zero Γ^k_{ij} for plane polar coordinates.

10. Use the covariant derivative $A^i_{;j} = \partial_j A^i + \Gamma^i_{jk}A^k$ to show that $A^i_{;i} = \dfrac{1}{\sqrt{g}}\,\partial_i\big(\sqrt{g}\,A^i\big)$, where $g = \det g_{ij}$, and use it to obtain $\nabla \cdot \mathbf{A}$ in spherical coordinates. (Here $\partial_j = \frac{\partial}{\partial q^j}$)

11. Consider polar coordinates $ds^2 = dr^2 + r^2\,d\theta^2$. Find the covariant derivative $V^\alpha{}_{;\alpha}$ of $\mathbf{V} = r^2\cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta$.

12. Using the definition of the covariant derivative, show that

$$A^\gamma{}_{;\alpha\beta} - A^\gamma{}_{;\beta\alpha} = -R^\gamma{}_{\delta\alpha\beta}\,A^\delta.$$